

Scalability & Future Plan

Date	03 July 2026
Team ID	
Project Name	Smart Lender Loan Eligibility Prediction System
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Limited concurrent user handling	May slow down under heavy traffic	High
2	Basic reporting module	Limited insights for stakeholders	Medium
3	Manual model retraining	Delays in adapting to new financial trends	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports ~500 concurrent users	Implement load balancing with Nginx + auto-scaling on IBM Cloud
2	Data Storage	MySQL database with limited indexing	Migrate to scalable cloud DB (PostgreSQL / MongoDB) with sharding
3	Performance	ML predictions optimized but logging slows under stress	Introduce caching (Redis), async logging, and optimized queries
4	Security	JWT authentication only	Add OAuth2, role-based access control, and encryption at rest

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Advanced Reporting & Analytics (PDF/Excel export, dashboards)	Q3 2026	Better insights for stakeholders
Phase 3	External API Integration (credit bureau, banking APIs)	Q4 2026	More accurate applicant profiling
Phase 4	Mobile App (Flutter-based) for applicants	Q1 2027	Wider accessibility and improved user experience
Phase 5	Automated Model Retraining (CI/CD for ML)	Q2 2027	Continuous improvement and adaptability